14.4, 14.5 Gradients, part 2

If $w = f(x,y,z)$, then we have f_x, f_y, and f_z. The partial derivative f_z is comput just like the others.

→ Do problem 1.

Directional derivatives are defined just as they are for functions of two variables:

$$\nabla f(x,y,z) = f_x(x,y,z)\vec{i} + f_y(x,y,z)\vec{j} + f_z(x,y,z)\vec{k}$$

and if $\vec{u}$ is a unit vector in $\mathbb{R}^3$, then

$$f_{\vec{u}}(x,y,z) = \nabla f(x,y,z) \cdot \vec{u}$$

→ Do problem 2.

<u>Facts</u> In $\mathbb{R}^3$, we have:

(1) ∇f points in the direction of greatest increase, and its magnitude is the maximum rate of change

(2) $\nabla f(a,b,c)$ is perpendicular (i.e., normal) to the level surface through (a,b,c)

(3) $f_{\vec{u}}(a,b,c) = 0$ iff $\vec{u}$ is tangent to the level surface through (a,b,c).

→ Do problems 3-6.

The tangent plane to the level surface of $f(x,y,z)$ through (a,b,c) is

$$f_x(a,b,c)(x-a) + f_y(a,b,c)(y-b) + f_z(a,b,c)(z-c) = 0.$$

→ Do problem 7.